1.0 Basic Concepts of Computer Networking

Introduction

A computer network is simply two or more computers connected together so they can exchange information. A small network can be as simple as two computers linked together by a single cable. This course introduces you to the hardware and software needed for a network, and explains how a small network is different from larger networks and the Internet. Most networks use hubs to connect computers together. A large network may connect thousands of computers and other devices together. A wireless network connects computers without a hub or network cables but use radio communications to send data between each other. Networking allows you to share resources among a group of computer users. If you have a printer connected to your computer, you can share the printer with other computers on the network. Then instead of buying a printer for every computer, all the computers can print across the network to the printer. If you already have access to the Internet from one computer on your network, you can share that Internet connection with other computers on the network. Then all the computers on your network can browse the Web at the same time, using this single Internet connection.

1.1 Computer Network

A computer network can be defined as a collection of computers and devices interconnected by communications channels that facilitate communications among users and allow users to share resources.

1.2 Advantages and Disadvantages of computer Networks

Advantages of Computer Networks:

i. Communication and Collaboration
ii. Resource Sharing
iii. Centralized Data Management
iv. Scalability
v. Remote Access
vi. Automation
vii. Enhanced Security
viii. Reduced Costs
ix. Improved Customer Service
x. Real-time Monitoring and Control

Disadvantages of Computer Networks:

i. Expensive
ii. Virus and Malware
iii. Management of the network
iv. Loss of Information
v. The system can be Hacked
vi. Lack of Privacy
vii. Security Difficulties
viii. Lack of Robustness
ix. Presence of Computer Viruses and Malware
x. Light Policing Usage Promotes Negative Ac

1.3 Types of Computer Networks

Local Area Network (LAN)

A local area network is a relatively smaller and privately owned network with maximum span of 10km to provide local connectivity within a building or small geographical area.

The LANs are distinguished from other kinds of networks by three characteristics:

(i) Size

(ii) Transmission technology, and

(iii) Topology

Metropolitan Area Network (MAN)

Metropolitan Area Network is defined as less than 50km and provides regional connectivity typically within a campus or small geographical area. It is designed to extend over an entire city. For example, a company can use a MAN to connect to the LANs in all of its offices throughout a city.

Wide Area Network (WAN)

Wide Area Network provides no limit of distance. A WAN provides long distance transmission of data, voice, image and video information over large geographical areas that may comprise a country, a continent or even the whole world.

1.4/1.5 Definitions of some Basic Networks Terms

Perimeter Networks: Perimeter networks are small networks that typically consist of only a few devices and are used to provide an additional layer of security between an organization's internal network and external networks like the internet.

Addressing: refers to the method used to identify devices on a network.

VLANs (Virtual Local Area Networks): VLANs are logical network segments that separate broadcast domains and improve network performance, security, and management.

Wired LANs: Wired Local Area Networks (LANs) use physical cables to connect devices within a limited geographical area, such as an office building or campus.

Wireless LANs: Wireless Local Area Networks (LANs) use radio waves to connect devices without the need for physical cables.

Leased lines: are private telecommunications circuits between two or more locations provided according to a commercial contract.

Dial-up: is a type of WAN connectivity that uses the Public Switched Telephone Network (PSTN) to establish a connection between two devices.

ISDN (Integrated Services Digital Network): is a type of digital telecommunications service that combines voice, video, and data services over a single circuit.

VPN (Virtual Private Network): is a secure network that uses encryption and other security measures to protect data transmitted over public networks.

T1 and T3: are digital transmission lines that provide high-speed data connectivity between two locations. T1 lines provide a data transfer rate of 1.544 Mbps, while T3 lines provide a data transfer rate of 44.736 Mbps

E1 and E3: are European equivalents of T1 and T3 lines, respectively. E1 lines provide a data transfer rate of 2.048 Mbps, while E3 lines provide a data transfer rate of 34.368 Mbps

DSL (Digital Subscriber Line): is a type of broadband internet connection that uses existing telephone lines to provide high-speed data connectivity.

Cable modem: is a type of broadband internet connection that uses cable television lines to provide high-speed data connectivity.

1.6 Different between client and server computers

- **Client Computer**: A client computer is a device used by end-users to access network resources and perform tasks related to their work. Clients initiate requests for services from servers and are typically smaller and less powerful than servers. **While**
- **Server Computer**: A server computer stores, processes, and distributes information for other computers on the network. Servers provide resources and services to clients, handling complex tasks like data storage, processing, and analysis. They are more powerful and have advanced hardware configurations compared to client computers.

1.7 Different between wired and wireless network

- Wired networks use physical cables such as copper wires, twisted pair, or fiber optic cables to connect devices. **while**
- Wireless networks use air as a medium to send electromagnetic or infrared waves.

Wired Networks	**Wireless Networks**
Uses physical cables to connect devices to the Internet or another network	Uses radio waves to connect devices to the Internet or anther networks
Devices are tethered to a router	Devices can roam untethered to any wires
Higher security due to physical access control	Lower security due to radio transmissions being available to anyone within range
Higher speed and bandwidth	Lower speed and bandwidth
Dedicated connection	Shared connection
More reliable due to the absence of interference	Less reliable due to interference from other networks or wireless-enabled devices
Higher installation cost due to physical cables	Lower installation cost due to the absence of physical cables
Higher maintenance cost due to physical cables	Lower maintenance cost due to the absence of physical cables

2.0 Functions of Hardware components of computer Networks

2.1 Hardware component of Computer Networks

The hardware components of a computer network include:

i. Network Interface Card (NIC)
ii. Hub
iii. Switch
iv. Router
v. Modem
vi. Server
vii. Transmission media
viii. Repeaters
ix. Gateway
x. Cables

2.2 Differentiate between Hub and Switch

Hub: a hub is a device that links multiple computers and devices together. Hubs can also be referred to as repeaters or concentrators, and they serve as the center of a local area network (LAN). While

Switch: A switch is a hardware component responsible for relaying data from networks to the destination endpoint through packet switching, MAC address identification, and a multiport bridge system.

S.No.	Hub	Switch
1.	Operates at the physical layer.	Operates at the data link layer.
2.	Does not support packet switching.	Supports packet switching and has separate collision domains.
3.	Follows broadcast transmission.	Supports multicast, unicast, and broadcast transmissions.
4.	Utilizes half-duplex transmission technique.	Utilizes full-duplex transmission technique.
5.	Does not support packet filtering.	Supports packet filtering.
6.	Typically has 4 ports.	Typically contains between 24 to 28 ports.

2.3 Networks Repeaters

A repeater is a networking device that amplifies and retransmits signals to extend the reach of a transmission. It is used in both wired and wireless networks to increase the network's range, restore a damaged or weak signal, or provide access to inaccessible nodes.

Functions of Repeaters

i. low cost
ii. improved network performance

iii. Enhanced signal strength
iv. Extended network distance.

2.4 Networks Bridges

A bridge in computer networks is a device used to connect multiple Local Area Networks (LANs) together with a larger LAN.

Functions Bridges

i. The ability to be used as a network extension.
ii. separate collision domains.
iii. increased bandwidth.
iv. high reliability and maintainability.
v. They can also divide the network into multiple LAN segments.
vi. control traffic in the network.
vii. interconnect two LANs with similar protocols.

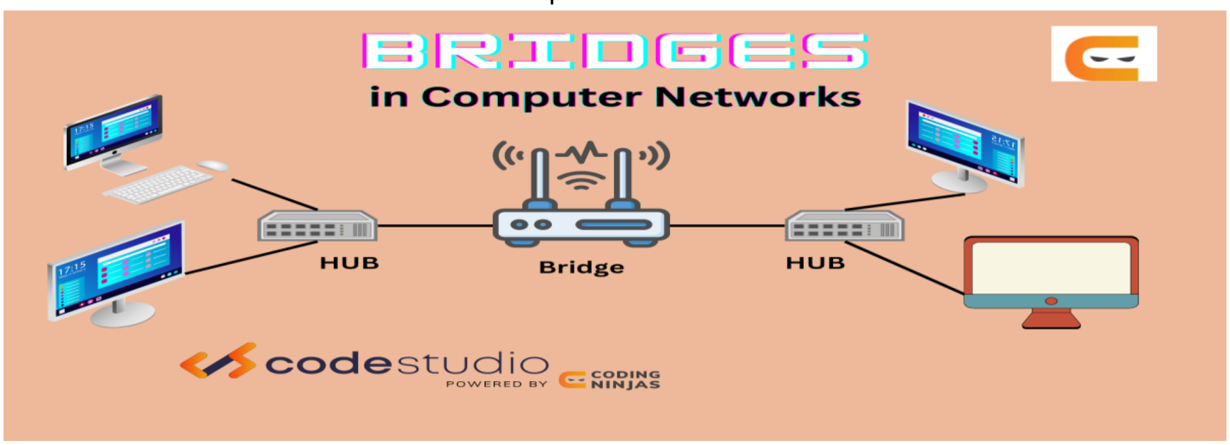

Fig 1. Bridges
Source: http://introductiontocomputernetwork

2.6 Networks Routers

Routers are essential devices in computer networks that receive, analyze, and forward data packets between different computer networks.

Functions of routers

i. Traffic Management.
ii. Connectivity.
iii. Routing Data.
iv. Security.
v. Network Expansion.
vi. Wireless Connectivity.

2.7 Network Interface Card (NIC): is a hardware component that connects a computer to a network, allowing for wired or wireless communication over the network.

Functions of NIC

i. Transforming computer data into a network-transmittable format.

ii. Assigning unique identifiers to data packets for source and destination identification.
iii. Regulating data flow, ensuring correct order, and minimizing errors during transmission.
iv. Detecting and resolving conflicts when multiple devices attempt to transmit simultaneously.
v. Filtering unwanted traffic, identifying/responding to threats, and implementing security features like firewalls and encryption for data protection.

3.0 Network Planning and Design

3.1 Definition

Network planning and design involve the strategic process of architecting a communications network to meet the needs of subscribers and operators. It encompasses topological design, network-synthesis, and network-realization, ensuring the network's reliability, scalability, and cost-effectiveness.

3.2 Importance of Network Planning and design

This section highlights the importance of network planning and design:

i. Equipment and Infrastructure Identification
ii. Operational Efficiency: Ensuring that operational needs are met without sacrificing efficiency.
iii. Improving network performance by determining optimal equipment numbers and avoiding bottlenecks.
iv. Planning for scalability to accommodate current and future organizational needs.
v. Incorporating security measures into network design, including access control and authentication procedures.
vi. Minimizing network interruptions through efficient design and operation.
vii. Ensuring network reliability through redundant components and backup systems.
viii. Minimizing costs by resource optimization and reducing the need for frequent upgrades.
ix. Ensuring network compliance with relevant regulations and standards, such as data privacy and accessibility.

3.3 Steps in designing a Network

The key steps in network planning and design include:

1. Topological design: Determining the placement of network components and how to connect them, using optimization methods from graph theory.
2. Network-synthesis: Determining the size of network components based on performance criteria like grade of service.
3. Network realization: Determining how to meet capacity requirements and ensure reliability, using multicommodity flow optimization

3.4 Network Topology and Access point

Network topology: refers to the physical and logical arrangement of the components in a network. It describes how the various nodes (devices) in a network are connected and how data flows between them.

Types of network topology

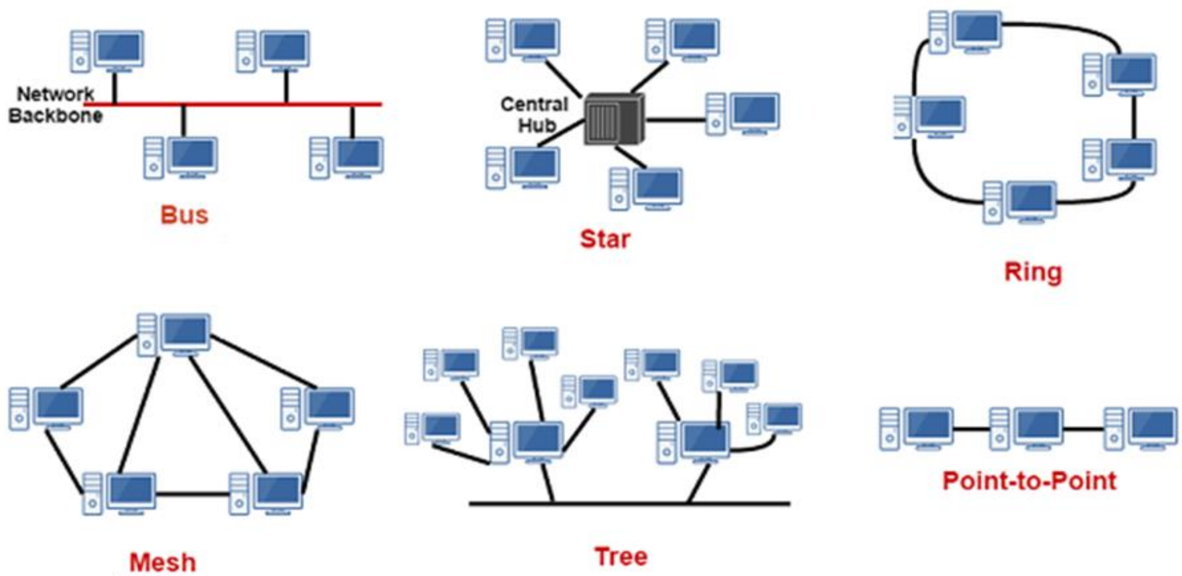

Fig. 2 Network Topology
Source: http://systemzone.net/networktopology

Network Access methods: refer to the techniques used to control access to the shared network medium and avoid collisions.

4.0 Network connections

4.1 Types of Network connections

A peer-to-peer network: is a decentralized network architecture where participants, known as peers, interact directly with each other without the need for a central server.

Point-to-point network: is a network topology that provides a dedicated connection between two endpoints, without any intermediary devices.

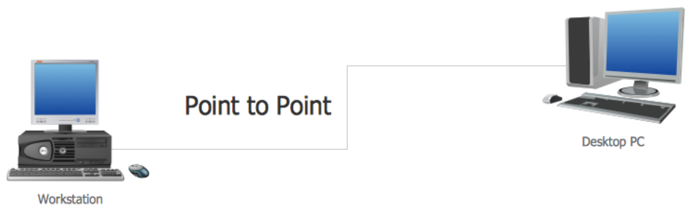

Fig. 3 Point to Point Network
Source: http://intoructiontocomputernetwork

A client-server network: is a distributed communications architecture where a centralized server receives and responds to requests for services and data from multiple clients.

4.2 Types of Cables termination

Termination Type	Suitable Cables
Heat Shrinkable Ends	Cables where heat can be applied to create a secure connection, providing protection against water, dust, and vibrations.
Cold Shrinkage Type	Cables where heat application is not feasible or desired.
Dry Wrap Type	Cables where a dry, non-invasive termination method is preferred.
Epoxy Resin Pouring Type	Cables that require a durable and protective seal against environmental factors.
Single Conductor Terminations	Cables with a single conductor, providing a secure connection for individual wires.
Multi-Conductor Terminations	Cables with multiple conductors, such as those used in power distribution or complex electrical systems.
High Voltage Terminations	High-voltage cables operating at 1kV, 10kV, 27.5kV, 35kV, 66kV, 110kV, 138kV, and 220kV, where insulation and safety are critical.

4.3 Advantages and disadvantages of each connection in 4.1 above

(i) point-to-point connection

Advantages of Point-to-Point Network	Disadvantages of Point-to-Point Network
Very simple and easy to configure and maintain	Limited to only two nodes, cannot be expanded

Advantages of Point-to-Point Network	Disadvantages of Point-to-Point Network
Dedicated bandwidth between the two nodes, with no sharing or contention	Entire network depends on a single link, if it fails the network goes down
Low latency and high data transfer speeds	Not scalable or flexible, adding more nodes requires additional equipment and wiring
Cost-effective for small networks with only two nodes	Lacks redundancy compared to other topologies

(ii) peer-to-peer connection

Advantages of Peer-to-Peer Networks	Disadvantages of Peer-to-Peer Networks
No single point of failure	Difficult to administer a decentralized system
Can easily scale by adding more peers	Viruses, spyware, and malware can easily spread
Peers contribute their own resources	Each computer needs its own backup system
Reduced infrastructure and operational costs	Used for transferring copyrighted files illegally
Direct communication and faster content delivery	Difficult to add new peers while maintaining performance

(iii) client server connection

Advantages	**Disadvantages**
Resources are centralized on servers, facilitating easier management and control	Requires powerful servers, high-capacity storage, and dedicated IT personnel
More clients can be easily added without significantly altering the existing infrastructure	If the server fails, the entire network can be affected and services become inaccessible
Services are more likely to be available as long as the servers are functioning	As the number of clients increases, demands on the server rise, potentially affecting its performance
Easier to manage permissions and access in a centralized server	Too many client requests can lead to network congestion and server overload

4.4 Types of Servers

i. **Print Servers:** A print server is a specialized server that manages and controls the printing process in a network environment.
ii. **Mail Servers:** A mail server is a computer system that processes and distributes email messages through a network.
iii. **Web Server:** A web server is responsible for publishing websites on the internet.
iv. **Database Server:** A database server stores and distributes databases via a network.
v. **Web Proxy Server:** A web proxy server acts as an intermediary between clients and web servers

4.5 Server Reliability, Availability and Data Integrity

Server Reliability: Reliability refers to the probability that a server will perform its intended function without failure over a specified time period under given conditions.
Server Availability: Availability refers to the percentage of time a server is operational and accessible to users or other systems.
Data Integrity: Data integrity ensures that information stored on servers is accurate, complete, and consistent over its entire lifecycle.

5.0 Open System Interconnection (OSI) Model & TCP/IP Reference Model

5.1 OSI model

OSI model is a standard model that defines a set of rules and requirements for data communication and interoperability between different devices, products, and software in a network infrastructure. The OSI model is a conceptual framework used to describe the functions of a networking system.

5.2 TCP/IP Reference Model

The TCP/IP (Transmission Control Protocol/Internet Protocol) Reference Model is a conceptual framework that describes the functions and protocols involved in data communication over a network.

5.3 Difference between TCP/IP and OSI Model

Parameter	TCP/IP Model	OSI Model
Number of Layers	4 layers: Application, Transport, Internet, Network Access	7 layers: Application, Presentation, Session, Transport, Network, Data Link, Physical

Parameter	TCP/IP Model	OSI Model
Approach	More horizontal, integrated approach with some overlapping between layers	Vertical, layered approach with clear boundaries between layers
Delivery Guarantee	Does not guarantee the delivery of data packets, relies on higher-level protocols for reliability	Guarantees the delivery of data packets
Replacement and Flexibility	More rigid, replacing protocols or adapting to new technologies is more challenging	Allows for easier replacement of protocols and adaptation to new technologies
Reliability	More widely adopted and proven to be more reliable in practical implementations	Considered less reliable than the TCP/IP model
History and Development	Developed by the ARPANET in the 1970s	Developed by the International Organization for Standardization (ISO) in the 1980s
Usage	The de facto standard for internet communication, widely used in modern networking	Primarily used as a reference model, not widely implemented in real-world networking

5.4 OSI layers and their functions

OSI Layer	Functions
Physical Layer	- Transmits message bits over a medium - Handles mechanical, electrical, procedural, and functional specifications for communication - Defines transmission mode, network topology, line configuration, data rate
Data Link Layer	- Breaks data packets into frames - Ensures error-free transmission of frames using MAC address - Provides flow control, framing, error control, physical addressing, access control
Network Layer	- Routes and forwards data packets between networks - Determines logical addressing and path selection for data transmission - Handles tasks like fragmentation and reassembly of data packets
Transport Layer	- Ensures reliable and error-free end-to-end data delivery

OSI Layer	Functions
	- Provides segmentation, flow control, and error correction mechanisms - Examples include TCP and UDP
Session Layer	- Establishes, maintains, and synchronizes communication sessions between applications - Provides checkpointing, recovery, and restart capabilities for long-running sessions - Handles tasks like authentication, reconnection, session termination
Presentation Layer	- Translates, formats, and encrypts/decrypts data for the application layer - Ensures data is in a format that applications can understand - Handles data compression, encryption, character encoding
Application Layer	- Serves as the interface for network services to applications - Defines protocols and standards for specific application-level tasks - Examples include HTTP, SMTP, FTP

6.0 IP Addresses on Networks using IPv4 and IPv6

6.1 Concept of IP Addressing and types

6.1.1 IP (Internet Protocol) ADDRESS

An IP address is a unique numerical identifier assigned to every device connected to a network, such as a computer, smartphone, or router.

6.1.2 Types of IP Addresses:

1. IPv4 (Internet Protocol version 4) Addresses:
    - IPv4 addresses are the most commonly used type, consisting of 32-bit numbers divided into four octets (e.g., 192.168.1.1).
    - IPv4 addresses can range from 0.0.0.0 to 255.255.255.255, providing approximately 4.3 billion unique addresses.
    - IPv4 addresses are further classified into different classes (A, B, C, D, and E) based on the range of the first octet.
2. IPv6 (Internet Protocol version 6) Addresses:
    - IPv6 addresses were developed to address the shortage of IPv4 addresses due to the growing number of internet-connected devices.
    - IPv6 addresses are 128-bit numbers, represented as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

- IPv6 addresses provide a much larger address space, allowing for a virtually unlimited number of unique addresses.

3. Public and Private IP Addresses:
   - Public IP addresses are assigned by Internet Service Providers (ISPs) and are used to identify devices on the public internet.
   - Private IP addresses are used within a local network, such as a home or office network, and are not directly accessible from the internet.
   - Private IP addresses are typically in the ranges 10.0.0.0/8, 172.16.0.0/12, or 192.168.0.0/16.
4. Static and Dynamic IP Addresses:
   - Static IP addresses remain the same over an extended period and are often used for services that require a consistent address, such as hosting a website or running a mail server.
   - Dynamic IP addresses are assigned by the ISP and change at set time intervals, providing a more cost-effective and flexible solution for most users.

6.2 Internet Protocol Version 4 (IPv4)

IPv4 is a foundational set of rules that allows devices like computers and phones to exchange data on the Internet. Each device connected to the Internet is assigned a unique IPv4 address, which is a 32-bit number written in decimal format with four decimals separated by periods. For example, an IPv4 address could look like 192.0.2.126. These addresses ensure that data is routed to the correct device on the network. IPv4 addresses are limited to approximately 4.3 billion unique addresses due to the 32-bit structure, leading to the exhaustion of available addresses.

6.3 & 6.4 Classes of IP Addresses and their ranges

1. **Class A:**
   - IP addresses in the range of 0.0.0.0 to 127.255.255.255
   - The first 8 bits (first octet) represent the network part, and the remaining 24 bits represent the host part.
   - Suitable for networks with a large number of hosts.
2. **Class B:**
   - IP addresses in the range of 128.0.0.0 to 191.255.255.255
   - The first 16 bits (first two octets) represent the network part, and the remaining 16 bits represent the host part.
   - Suitable for medium-sized networks.
3. **Class C:**
   - IP addresses in the range of 192.0.0.0 to 223.255.255.255
   - The first 24 bits (first three octets) represent the network part, and the remaining 8 bits represent the host part.
   - Suitable for small networks with a limited number of hosts.
4. **Class D:**
   - IP addresses in the range of 224.0.0.0 to 239.255.255.255

- Used for multicast addressing, where a single packet can be delivered to a group of devices.

5. **Class E:**
    - IP addresses in the range of 240.0.0.0 to 255.255.255.255
    - Reserved for future use and experimental purposes.

6.5 VLSM (Variable Length Subnet Mask)

VLSM is a technique used in IP network design to create subnets with different subnet masks.

6.6 Internet Protocol version 6 (IPV6)

IPv6: is the latest version of the Internet Protocol, which is the primary protocol used to transmit data over the internet. IPV6 was developed to address the limitations and eventual exhaustion of the previous version, IPV4 (Internet Protocol version 4).

6.7 Network functionality testing

Network functionality testing: refers to the process of evaluating the performance, reliability, and overall functionality of a network infrastructure. The goal of network functionality testing is to ensure that the network is operating as intended and meeting the required specifications and expectations.

7.0 Wireless Network Access

7.1 Different between internet and extranet

Parameter	**Internet**	**Extranet**
Usage	Public	Private
User Types	Any user with Internet access	Suppliers, customers, business partners
Usage	Access all kinds of information	Check status of orders, access data, send emails
Security	Low security, configured under 0 security level in firewall	Generally, uses VPN technology for secured communication over the Internet, medium security level
Regulated by	Internet Architecture Board (IAB), Internet Corporation for Assigned Names and Numbers (ICANN)	Multiple organizations
Coverage	Wide area	Within an organization
Access	Large number of users	Limited number of users

7.2 TYPES OF INTERNET CONNECTIVITY

1. Dial-Up Connection:
    - Established between the computer and the ISP server using a modem over a telephone line.
    - Provides a slow and traditional internet connection, not preferred nowadays.
    - Requires dialing a phone number to access the internet, and the telephone line cannot be used for other purposes during the connection.
2. Broadband Connection:
    - Refers to high-speed internet access that is faster than traditional dial-up.
    - Includes technologies like DSL, cable, fiber-optic, and wireless connections.
    - Allows simultaneous use of the internet and telephone services.
    - Provides a wide bandwidth for data transmission.
3. DSL (Digital Subscriber Line):
    - Provides an internet connection through the telephone line network.
    - Always-on connection, no need to dial a phone number.
    - Speeds range from 128 Kbps to 8 Mbps, depending on the service.
4. Cable Internet:
    - Provides internet access through the cable TV infrastructure.
    - Uses a cable modem to connect to the internet.
    - Offers fast download and upload speeds, ranging from 512 Kbps to 20 Mbps.
5. Fiber-Optic Internet:
    - Uses fiber-optic cables to transmit data at the speed of light.
    - Offers the fastest internet speeds, typically ranging from 200 Mbps to 2 Gbps.
    - Provides symmetrical download and upload speeds.
    - However, it has limited availability compared to other connection types.
6. Satellite Internet:
    - Accesses the internet via a satellite in Earth's orbit.
    - Suitable for areas where broadband connections are not available.
    - Speeds range from 512 Kbps to 2 Mbps, with higher latency due to the long signal travel distance.
7. Wireless Internet:
    - Uses radio frequency bands to connect to the internet without telephone lines or cables.
    - Provides an always-on, mobile internet connection.
    - Speeds range from 5 Mbps to 20 Mbps, depending on the technology used.
8. Cellular Internet (3G/4G/5G):
    - Provides wireless internet access through cellular networks.
    - 3G networks offer speeds around 2 Mbps, while 4G networks can reach up to 21 Mbps.
    - 5G networks aim to achieve peak speeds of 100 Mbps or higher.

7.3 Wireless Network

A wireless network is a type of computer network that uses radio waves to transmit data between devices, rather than relying on physical cables. Wireless networks allow devices to connect and communicate with each other without the need for a wired infrastructure.

Types of Wireless Access

i. Wireless Local Area Network (WLAN): WLANs provide internet access and network connectivity within a limited geographical area, such as a home, office, or small business.
ii. Wireless Metropolitan Area Network (WMAN): WMANs cover a wider geographical area, such as a city or metropolitan region, providing wireless access over a larger scale.
iii. Wireless Wide Area Network (WWAN): WWANs provide wireless connectivity over a wide geographical area, often on a national or global scale.
iv. Wireless Personal Area Network (WPAN): WPANs connect devices within a personal space, typically within a range of 10 meters or less. E.g Bluetooth

7.4 Difference between Dial-up, wireless and broadband internet access

Parameter	Dial-up	Wireless	Broadband
Speed	Slow, typically up to 56 Kbps	Speeds vary, ranging from 5 Mbps to 20 Mbps	Faster speeds, typically ranging from 700 Kbps to several hundred Mbps
Connection	Uses a modem over a telephone line	Uses radio frequency bands for connectivity	Uses various technologies like DSL, cable, fiber, and satellite
Mobility	Limited mobility due to physical connection	Provides mobility and flexibility	Offers both stationary and mobile connectivity
Availability	Widely available but declining in popularity	Widely available in urban areas, limited in rural areas	Widely available in urban areas, limited in rural areas
Usage	Suitable for basic internet browsing and email	Suitable for general internet use and some streaming	Suitable for high-speed internet activities like streaming, gaming, and large downloads
Cost	Relatively inexpensive	Costs vary depending on the technology and service provider	Costs vary but generally more

7.5 Advantages of Broad band Over Dial-up and Wireless Access Network

Advantage	**Broadband over Dial-up**	**Broadband over Wireless Access**
Speed	Significantly faster download and upload speeds, typically ranging from 700 Kbps to several hundred Mbps, compared to dial-up's 56 Kbps.	Generally, offers higher download and upload speeds compared to wireless access, which can vary based on the technology used.
Reliability	More reliable and stable connection, with no interruptions during use, unlike dial-up which can be disrupted by phone calls.	More reliable and less prone to disruptions caused by weather or other environmental factors that can affect wireless signals.
Simultaneous Usage	Allows simultaneous use of the internet and telephone, without tying up the phone line.	N/A (not a direct advantage over wireless)
Always-on Connectivity	Provides an "always-on" connection, eliminating the need to dial-up and disconnect each time.	N/A (not a direct advantage over wireless)
Suitability for High-Bandwidth Applications	Better suited for bandwidth-intensive activities like streaming, online gaming, and large file downloads/uploads.	Better suited for high-bandwidth applications compared to some wireless access, especially in areas with limited coverage or capacity.
Security	When properly secured, can offer better protection against cyber threats compared to some wireless access networks.	N/A (not a direct advantage over wireless)
Availability	More widely available, especially in urban and suburban areas.	May have limited availability in remote or rural areas.
Cost	Generally, more expensive than dial-up.	Generally, more expensive than wireless access.

7.7 Wireless network standard

Wireless network standards refer to the technical specifications and protocols that define the operation and interoperability of wireless communication devices and networks.

The main IEEE 802.11 wireless standards include:

1. 802.11a (Wi-Fi 2): Introduced in 1999, this standard operates in the 5 GHz frequency band and supports up to 54 Mbps data rates.
2. 802.11b (Wi-Fi 1): Introduced in 1999, this standard operates in the 2.4 GHz frequency band and supports up to 11 Mbps data rates.
3. 802.11g (Wi-Fi 3): Introduced in 2003, this standard also operates in the 2.4 GHz band but supports up to 54 Mbps data rates.
4. 802.11n (Wi-Fi 4): Introduced in 2009, this standard supports both 2.4 GHz and 5 GHz bands and can achieve data rates up to 600 Mbps using MIMO (multiple-input, multiple-output) technology.
5. 802.11ac (Wi-Fi 5): Introduced in 2013, this standard operates in the 5 GHz band and can achieve data rates up to 1.3 Gbps.
6. 802.11ax (Wi-Fi 6): The latest standard, introduced in 2019, operates in both 2.4 GHz and 5 GHz bands and can achieve data rates up to 9.6 Gbps.

7.7 Types of network security

i. Access Control: Controls which users have access to the network or specific sensitive sections.
ii. Antivirus and Anti-Malware Software: Protects against malware, including viruses, worms, Trojans, ransomware, and spyware.
iii. Application Security: Focuses on securing software products and devices within the network environment.
iv. Behavioral Analytics: Establishes a baseline of normal behavior for users, applications, and network traffic.
v. Firewalls: Monitors incoming and outgoing network traffic and enforces security rules.
vi. Intrusion Prevention Systems (IPS): Scans network traffic to actively block attacks.
vii. Network Segmentation: Defines boundaries between network segments based on function, risk, or role.